

Article

Carbon Accounting and Beyond: An Evidence-Based Life Cycle Assessment of the Environmental Impacts of Data Center IT Equipment

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Abstract

Data centers are essential to modern infrastructure, but are significant contributors to greenhouse gas (GHG) and related environmental challenges. Despite energy efficiency improvements, rising electricity demands driven by technologies like AI pose challenges to sustaining low-emission sector growth. Regulatory requirements mandate data center operators to report electricity use and emissions, yet current methods, particularly pertaining to indirect sources, remain insufficient. This study explores how life cycle assessment (LCA)-based estimates of data center IT equipment impacts compared with commonly used corporate GHG accounting methods: average-data and spend-based. Environmental impacts were modeled at both grouped and granular levels to support enterprise-wide reporting and operational decision-making. Sensitivity and uncertainty analysis validate the robustness of the LCA models. Scenario analysis was also conducted to assess emission abatement strategies, including on-site renewable energy generation and operation with a low-carbon electricity grid. The results show that the LCA approach produced emissions of 0.710 CO₂ eq/kWh, along with additional burdens that are not captured through carbon-only metrics. The LCA results are closely aligned with the average-data method (0.723 kg CO₂ eq/kWh) while the spend-based method yielded substantially higher estimates (1.07 kg CO₂ eq/kWh), highlighting the inaccuracies associated with volatile market prices. Scenario analysis identifies grid decarbonization as the most effective mitigation pathway, while also demonstrating environmental trade-offs across impact categories. Findings highlight the importance of comprehensive LCA-based assessment methods to improve emission reporting accuracy, transparency, and sustainability-focused decision-making in the data center sector.



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1. Introduction

Data centers are facilities that provide shared access to data and applications using a complex network, computer, and storage infrastructure, and act as a critical resource to nearly every business, government, and university entity. The International Energy Agency (IEA) estimated that data centers accounted for 330 metric tons of carbon dioxide equivalent (CO₂ eq) in 2020, or 0.6% of overall greenhouse gas (GHG) emissions [1]. In

2021, data centers and data transmission networks together accounted for approximately 1–1.3% of global electricity demand [1]. While demand for data center services has grown substantially over the past decade, improvements in hardware, infrastructure, and energy transmission have largely negated proportional GHG emission increases [1]. Despite efficiency gains, continued industry growth and the rise in computing needs associated with artificial intelligence (AI) applications and cryptocurrencies are likely to lead to an increase in energy use and related emissions. The IEA notes that growth in workloads between 2015 and 2023 led to a 20–40% increase in data center energy usage. It predicts the electricity consumption of data centers, along with AI and cryptocurrencies, to range between 620 and 1050 terawatt-hours (TWh) in 2026, which represents an additional 160–590 TWh electricity compared to 2022 [1].

Due to the significant energy demand from data centers, recent regulatory actions require data center operators to report energy use and emissions to promote transparency and accountability related to electricity consumption and management. For example, the European Union (EU) introduced mandatory reporting in 2024 which requires data center operators to report their energy and emissions and setting climate neutrality goals by 2030 as part of the Energy Efficiency Directive [2]. However, calculating the GHG emissions associated with data centers is not straightforward. In addition to the direct and indirect emissions associated with energy use such as electricity and the cooling of data centers, there are a number of other indirect sources of emissions from the value chain such as the computing and building infrastructure systems housed within a data center. However, emission calculation and reporting methods are lagging behind as data center operators continue to use approaches to calculate carbon emissions that either overlook or estimate embodied value chain/indirect emissions [3]. Accordingly, there have been calls for a science-led transition towards more reliable GHG emission calculation and estimation [4]. Further, many emissions accounting methods including the GHG corporate standards overlook other environmental hotspots such as pollutants and resource scarcity associated with the life cycle of the components and processes related to the data center.

The proliferation of data centers and the concern regarding their ecological impact has led to growing—though still limited—academic attention towards assessing their environmental impact. Several studies have addressed the environmental impact of data centers (e.g., [5–10]). However, much of the work is based on reference models of data centers, rather than empirical data from operational facilities. Further, many of these studies limit their assessment to a single metric (e.g., GHG emissions) rather than a more comprehensive environmental impact assessment, giving an incomplete outlook on overall environmental impact when analyzing a resource-intensive sector [5–7]. In addition, previous work has focused on data centers operating within specific geographies, making the results not necessarily applicable to the United States (US) context [7,9,11,12]. To our knowledge, no studies have produced results that allow for a granular analysis of data center equipment that would allow insight to design improvements. Furthermore, previous work has largely concentrated on estimating the impact, with limited attention given to comparing assessment approaches or offering recommendations for improving evaluation methods.

1.1. GHG Protocol Standards

The GHG corporate and corporate value chain (Scope 3) standards offered by the GHG Protocol are the most widely used GHG accounting standards, and organize emissions into three categories—Scope 1, Scope 2, and Scope 3—as a means to measure, calculate, estimate, and report GHG emissions. Accordingly, most leading companies in the ICT (Information and Communications Technology) sector such as Amazon and Microsoft use the GHG Protocol to quantify organizational carbon emissions. Scope 1 covers an organization's

direct/on-site emissions from fuel combustion (such as diesel, gasoline) and refrigerants in data centers. Scope 2 covers indirect emissions from purchased electricity, steam, heating and cooling by the organization. Scope 2 emissions are particularly important for data centers given the significant electricity and cooling requirements. Scope 3 covers all other indirect emissions from their value chain. Specifically, Scope 3 emission sources include upstream and downstream activities, such as the procurement of products, services and capital goods; upstream emissions from fuel and energy-related activities; and end-of-life treatment of products. The GHG Protocol's Scope 3 standard outlines 15 categories (upstream and downstream), but not all categories are relevant to a data center. For example, downstream transportation and distribution is not a relevant category considering the companies who own or operate data centers (e.g., Equinix, NTT Data) do not produce or sell physical goods that may require physical transportation prior to resale. However, emissions from procurement of underlying information technology (IT) infrastructure can be the source of a significant portion of embodied value chain emissions (Scopes 2 and 3) within a data center.

The GHG corporate standards summarize common methods for calculating emissions (in order of accuracy): (1) supplier-specific, where product-level primary data is collected directly from suppliers; and (2) hybrid, which is a combination of primary and secondary data. In the case of Scope 3 emissions associated with purchased goods and services (category 1) and capital goods (category 2), it lists two additional and relatively less accurate approaches; (3) average-data, where emissions are estimated based on relevant units (i.e., mass) of goods/services and average emissions per unit; and (4) spend-based, where emissions are estimated based on the financial value of goods/services and average emissions per unit. When applying any of these methods, the measurement of Scopes 1 and 2 emissions using supplier-specific data is relatively straightforward as compared to Scope 3, considering the primary source data for Scope 1 and 2 emission sources are likely to be more accessible (e.g., emissions associated with use of specific amounts of fuels used on-site, and electricity and cooling procured). However, Scope 3 or value chain emissions can be difficult to measure and regulate as they are generated by entities external to the reporting organization, with limited visibility and control [6,13,14].

1.2. Limitations of the GHG Corporate Standards for Environmental Impact Assessment

There are two primary deficiencies in the current utilization of the GHG corporate standards for environmental impact evaluation. First, there are a number of challenges associated with estimating Scope 3 indirect emissions particularly due to the complexity in capturing supplier-specific primary source data [6,14,15]. For example, a data center contains complex IT systems with thousands of underlying components and equipment which have embodied carbon from the production, transport, and end-of-life processes. To quantify emissions associated with the equipment and underlying components requires inputs from the upstream and downstream supply chain partners that design, manufacture, and dispose of these devices. However, most manufacturers of IT equipment do not provide product-specific emissions data for their product portfolio due to the complexity and resource-intensive nature of such an assessment. Given this challenge, most data center companies utilize either the average or spend-based methods to calculate emissions associated with category 1 and 2 of Scope 3 emissions which often represent the largest share of emissions.

Specifically, the spend-based method, which as noted by GHG calculation guidance is the least specific to a company's actual activities thereby rendering low-quality emissions data, is often used to determine the indirect emissions associated with Scope 3 categories as the economic value of goods/services is captured in accounting and procurement systems

and is easily accessible [16,17]. Further, Scope 3 emissions are calculated at the corporate level and disaggregating the corporate emissions for specific data centers can be challenging. As an example, due to centralized procurement, departments may not track purchases of equipment (purchased goods and services and capital assets) at the data center level which can make it difficult to allocate emissions associated with these purchases to a specific data center, particularly when a company has a portfolio of hundreds of data centers under various operational and ownership arrangements (e.g., serviced, leased, owned, etc.). Many large data center companies such as Amazon and Microsoft do not break down the emissions corresponding to the data center portion of their business. Therefore, the emissions footprint associated with data centers is often unclear.

Additionally, data center service providers often have unique business models wherein some provide colocation services by offering space, power and cooling only, and multiple customers/tenants bring and manage their own IT equipment (e.g., Equinix, CyrusOne Inc., NTT Data). Others provide cloud services (e.g., AWS, Microsoft Azure) which fall under the owner-operated category and offer both space, power, cooling, and IT equipment [18]. Accordingly, the indirect emission footprint can vary significantly as per the underlying business models and organizational boundaries. A survey of CDP (Carbon Disclosure Project) disclosure of various companies in the data center space is indicative of the differences in the scope and scale of the Scope 3 emissions reported. For example, as per the 2023 CDP disclosure, Cyrus One Scope 3 emissions were 19% of total emissions, 35% in the case of Equinix, and 91% for NTT Data. There are deficiencies in the quality and completeness of data reported across corporations particularly when it is not audited and assured; it is estimated that only 18% of companies report Scope 3 emissions [4,6].

Second, the GHG corporate standards can be used to assess environmental impact associated with GHG emissions. Although for data centers, operational electricity demand is one of the most significant emission sources, the environmental impact of the value chain such as water consumption and the release of other pollutants and toxins in the environment in the production and use of IT equipment can be material. Given the focus on GHG emissions, the environmental impact of the production and disposal of these components is overlooked. For example, byproducts of the IT hardware manufacturing process such as sulfidic tailings have a significant environmental impact due to the release of heavy metals and carcinogens in the environment [3,9]. Aggregated GHG accounting may also obscure intervention points; studies have shown that emissions depend heavily on material and process design, and more rigorous environmental impact analysis techniques are needed to capture these “hotspots” to identify potential areas for intervention [3,19,20]. Therefore, it is important to note that impact of value chain actions such as the frequency of IT refresh also play a significant role in managing the overall environmental impact of data centers. Given these gaps, improved approaches are warranted to better understand the environmental impact of data centers. We propose that using a life cycle assessment (LCA) approach can offer a solution to address some of the gaps and challenges noted above.

Furthermore, nearly all prior studies aim solely to quantify environmental impact rather than evaluate the performance of different assessment methodologies. To date, no published work has systematically compared equipment-level LCA results with corporate GHG accounting approaches. This gap is notable because data centers commonly rely on spend-based Scope 3 calculations, which are known to introduce substantial uncertainty and often fail to reflect the material intensity of specific equipment types. A methodological comparison is therefore essential to understand the misalignment between procurement-based accounting and physical life cycle impacts, and to guide data center managers in selecting appropriate tools for sustainability decision-making. This research speaks to the benefit of applying the LCA technique of environmental accounting as a comprehensive

method which looks beyond single-scope emissions (i.e., solely focusing on CO₂ equivalents) as well as accounting for the emissions bled through each stage of the life cycle in order to guide data-informed decisions. By simplifying emissions accounting to solely review CO₂, data centers and other industries miss the opportunity for sustainability action improvements which could be addressed throughout the supply chain and operational life cycle [3,21].

1.3. Life Cycle Assessment (LCA) for Improved GHG Reporting

LCA allows for the combination of all four methods of data collection outlined by the GHG corporate standards [4]. However, the LCA approach focuses on disaggregating data as a way to look at each individual component within the life cycle system boundary, allowing for a more introspective analysis of GHG emissions and overall environmental impact. While LCA may need to rely on estimations and calculations in order to disaggregate data, it accounts for risk or uncertainty in the model which improves the level of transparency. By acknowledging and including level of uncertainty in the model, LCA interpretation can more readily identify next steps for analytic improvement. This can act as a call to action for obtaining additional primary data, as well as identify the priority level for certain sources of emissions data. Furberg et al. (2026) recently published assessment guidelines for the ICT sector's direct energy use and climate impact, noting the complexity of this effort due to the varied nature of ICT business design, services provided, and relatively short lifetime of IT equipment. The authors call for greater standardization to improve future projections of climate impacts for proactive planning and decision-making surrounding the ever-growing sector [22,23]. By creating a framework for data center IT equipment LCAs, this study provides a foundation to move away from less reliable secondary-data-centric methods to improve carbon emission calculation and reporting accuracy, as well as data reporting transparency.

LCA is a comprehensive approach to environmental emissions accounting, which considers up- and downstream activities throughout the entire life cycle of a product or process. LCA was first proposed in the 1960s, and has undergone robust development in both its application and methodologies. LCA stands apart from other GHG emissions accounting techniques, calling for rigorous standardization following ISO (International Organization for Standardization) 14040 and 14044 for principles and framework, and requirements and guidelines, respectively [24,25]. Other environmental accounting techniques, such as input–output analysis, environmental impact assessments, material flow analysis, and ecological footprint assessment, are generally less comprehensive and allow for greater assumptions and estimations in their models. Within the LCA approach, there are several methodologies and frameworks which allow for different estimations, visualizations, and categorizations of emissions beyond carbon and include other key pollutants as well as primary areas of pollution concern (i.e., ground vs. water vs. air) [26]. Calculating emissions and defining how they may disproportionately impact certain areas of the environment supports informed decision-making not only to reduce overall emissions, but also to focus on environmental hotspots impacted by the product or process being studied. Further, LCA is now being applied to sustainable development modeling, providing real-time results in early design stages and scenario analysis to inform science-based sustainable management decisions [27–29].

The rapid expansion of data centers has elevated concern over their environmental impact, prompting a growing but still limited body of research assessing their sustainability performance. As global digitization accelerates, data centers have become essential infrastructure for cloud computing, AI, and high-volume data processing, yet their rising electricity demand, embodied material use, and water consumption pose substantial sus-

tainability challenges. Whitehead et al. (2014, 2015) published a two-part report assessing the environmental impacts of data centers, that details data center electricity consumption, the challenges associated with using single-issue metrics, and the application of environmental assessment methods for this industry [8,9]. They argue for the need to apply life cycle thinking as a means to discontinue using “operational efficiency as a proxy for sustainability” in order to avoid the issue of overlooking how changes in one stage of business may impact another. There have been several attempts to quantify emissions associated with data center technology and operations using LCA and other environmental accounting methods [5,6,9–12,30]. Much of the existing literature focuses primarily on operational electricity consumption, often using high-level carbon footprinting or top-down energy modeling to estimate GHG emissions [5,6,31,32]. While these studies have improved understanding of the sector’s energy use, their single-metric scope and reliance on simplified assumptions provide only a partial view of environmental impacts.

Comprehensive LCA studies remain comparatively scarce. Existing LCAs of data centers tend to rely on reference facility models or aggregated component assumptions rather than empirical, equipment-level data from operational sites [7–10]. As a result, they offer limited insight into the material and resource burdens embedded in IT hardware which are responsible for substantial upstream impacts. This omission is consequential, as data centers experience rapid IT equipment turnover driven by performance scaling, reliability requirements, and increasingly AI-oriented workloads. Recent assessments highlight that embodied impacts from IT hardware can rival operational impacts, depending on refresh cycles and local electricity mixes [9,11]. More recently, a high-profile cradle-to-grave LCA by Microsoft/WSP (2025) compared cooling strategies at cloud scale, showcasing the power of LCA as a means to inform engineering design and reduce environmental impact, while also informing regulation and policy by enabling the evaluation of trade-offs [33]. This work also highlighted significant data gaps in component-level inventory data [33]. Therefore, data center managers require tools that account for both electricity-related impacts and the resource burdens on continual equipment replacement, particularly as supply chain constraints and circularity objectives gain strategic importance. The details of this limited literature review are summarized in Table 1.

Table 1. Summary of relevant environmental impact studies.

Study Focus	Functional Unit	Accounting Method	Key Limitations	Geography	Source
Develops a GHG inventory and forward-looking scenario analysis for data centers, modeling energy use and emissions (Scope 1–3) and evaluating pathways to climate neutrality by 2030.	Per data center.	GHG Protocol-based accounting (Scope 1, 2, and 3) using a hybrid bottom-up/top-down model.	Simplified and non-representative model (single data center, fixed device types); excludes several Scope 3 categories, relies on assumptions and limited data transparency.	Germany/EU	[6]
Evaluates the global warming potential (GWP) of a theoretical data center using LCA, comparing multiple scenarios that vary energy mix, IT refresh rates, and physical infrastructure to identify emission-reduction strategies.	Per data center.	LCA	Based on a theoretical case with secondary data, simplified assumptions, and focus on a single impact category (GWP).	Italy/EU	[7]

Table 1. Cont.

Study Focus	Functional Unit	Accounting Method	Key Limitations	Geography	Source
Compares environmental impacts of solar-powered edge data centers with refurbished servers versus conventional grid-powered data centers using LCA across multiple design scenarios.	Defined as “providing IT hosting services for five servers and one router over a 15-year operational period”.	LCA	Relies on assumed system configurations and simplified modeling while excluding broader system impacts such as networks, user devices, and operational variability.	France	[10]
Examines the environmental impacts of a Tier 4 data center using LCA, focusing on operational activities such as electricity, cooling, fuel use, and material inputs in a real-world case study.	Defined as “the annual operation of the data center”.	LCA	Limited comparability due to a unique functional unit, reliance on a single facility’s data, partial exclusion of transportation modeling details.	Turkey	[11]
Analyzes the environmental footprint of the data center service sector and develops a multi-criteria framework to evaluate optimal site locations based on energy, water, and environmental factors.	Not a single fixed unit.	Employs a hybrid approach combining LCA-based emission factors with energy–water–GHG calculations.	Relies on heterogeneous data sources with varying accuracy and aggregates results at regional levels.	Sweden	[12]
Evaluates the accuracy and quality of corporate carbon disclosures, comparing GHG emission reporting in annual reports versus CDP submissions across scopes 1, 2, and 3.	Not a single fixed unit.	Uses a comparative disclosure assessment framework based on GHG Protocol categories.	Finds issues of inconsistent reporting, incomplete scope 3 disclosures, overreliance on market-based instruments (e.g., RECs), and lack of regulatory standardization and verification.	Global	[15]
Develops a methodology to model the energy consumption and GHG emissions of electronic software distribution (ESD) and compares it with physical distribution alternatives (e.g., DVDs).	Defined as “the purchasing and distribution of one software program from an online store”.	LCA	Includes significant uncertainties due to assumptions on server utilization, network infrastructure, device characteristics, and data confidentiality, as well as variability in LCA boundaries and parameters.	UK	[21]
Evaluates the environmental impacts of a data center cooling device, focusing on the manufacturing phase and comparing different production scenarios (e.g., electricity sources, waste reduction, forklift technologies).	Per data center cooling unit.	LCA	Focused mainly on the manufacturing phase, not the full life cycle and based on specific regional data.	Turkey	[30]

Table 1. *Cont.*

Study Focus	Functional Unit	Accounting Method	Key Limitations	Geography	Source
Analyzes the carbon footprint of computing systems (ICT) across mobile devices and data centers, focusing on life cycle emissions and highlighting the shift from operational energy to hardware manufacturing impacts.	Not a single fixed unit.	Carbon footprinting	Relies on public industry data that may be incomplete or inconsistent, and focuses mainly on carbon emissions, excluding other impacts (e.g., water, toxicity).	Global	[31]
Evaluates the life cycle carbon footprint and electricity use of the entire ICT system in Sweden, including user devices, networks, data centers, and operator activities.	One year of ICT network operation.	LCA	The analysis is limited by uncertainties in data allocation, reliance on variable secondary data for user equipment, and a focus on carbon and energy rather than broader environmental impact.	Sweden	[32]
Evaluates the life cycle environmental impacts of different data center cooling technologies within cloud infrastructure, including servers, cooling systems, buildings, and energy systems.	Annualized virtual core units.	LCA	Analysis is limited by data gaps requiring proxies (especially for cooling fluids and electronics), uncertainty in inventory data, and challenges in modeling complex cloud system interactions.	US	[33]

1.4. Conceptual Framework and Study Objectives

Despite these widespread challenges in addressing environmental accounting for data centers, IT product-level emissions data required for the supplier-specific and hybrid methods is not widely available. This limited transparency often forces data center managers to rely on the less reliable approaches such as using the spend-based method to estimate indirect value chain emissions. To address these gaps and highlight the advantages of more reliable emissions estimate methods, we demonstrate an LCA model-based approach that addresses both direct and indirect emissions from the IT equipment used in a data center enterprise, and compare the results to GHG corporate accounting approaches using average-based and spend-based data. We specifically focus on IT equipment considering that indirect emissions from the production and use of IT equipment has been highlighted in the literature to be one of largest contributors of environmental impact of a data center [7,9,11]. The central hypothesis of this study is that LCA can be used to improve the accuracy of carbon emission calculation and reporting, and to better understand and manage the environmental hotspots related to the production and use of IT equipment in data centers.

To this end, we first develop a comprehensive system boundary and create a life cycle inventory. We then apply the framework through an empirical case study to calculate the lifetime environmental impacts. We validate the robustness of the model using sensitivity analysis to evaluate variability of IT equipment electricity consumption, and uncertainty analysis to identify areas of risk and model limitations. We then apply scenario analysis to explore two potential on-site and off-site energy generation scenarios to evaluate alternate possibilities to reduce overall environmental impact, given the significant energy demand

associate with data centers. Finally, we calculate emissions using the GHG-corporate-standards-based accounting approach and the average-data and spend-based methods to empirically compare the relative accuracy and completeness of these more commonly used approaches with the LCA-model-based approach.

In the following sections, we detail the methodologies used to estimate life cycle emissions, followed by the results of a case study demonstrating the model's real-world application. This is accompanied by a discussion comparing our findings to those of similar studies, validating the approach. Finally, we provide recommendations for data center managers on integrating this framework into emission reporting to enhance science-based decision-making and reduce environmental impacts.

2. Methodology

2.1. Life Cycle Assessment

We perform the LCA using SimaPro Version 10.2 software because it contains several impact assessment methods and an extensive inventory of databases that we modified to best conform to the parameters of our analysis for our study area. We use the ReCiPe 2016 framework, which is well suited for scenarios using a global value chain. In this study, we use both midpoint and endpoint methods, using the hierarchist perspective for the analysis (see Table S1), which is the consensus model most commonly used in scientific research [34,35]. The midpoint method is suitable for detecting environmental impacts early in the cause–effect chain, and the calculated emissions are based on a global perspective on a 100-year time horizon; the endpoint method is better suited to evaluate the environmental impact at the end of the cause–effect chain and is based on damages derived from the midpoint characterization factors [34]. By showcasing both midpoint and endpoint indicators, we show the disaggregated (midpoint) and aggregated (endpoint) emission indicators which can be useful for result communication and interpretation.

Contribution analysis in LCA allows us to determine which processes play a significant role in the results in the graphical form of a process tree, or Sankey diagram. Inventory analysis provides a list of substance emissions for the midpoint and endpoint impact categories, and calculates the emissions associated with each of the impact categories. Sensitivity analysis allows us to test the reliability of study results by examining how variations in input parameters affect environmental outcomes. Uncertainty analysis quantifies the confidence in LCA results by addressing data variability. Scenario analysis allows us to relate the substance emissions of multiple processes. Through these multiple analytical pathways, we gain a more accurate depiction of the environmental impact of a system or systems, providing evidence-based results which may inform management, strategy, and policy decisions.

Further, we present the results using a dual approach: granular and grouped. While the net results for both approaches are the same, the presentation of the data differs. The granular model shows a breakdown of each generation of equipment and its associated electricity use. This approach is most appropriate for data center managers as the granular information can help to guide decisions on transitioning to newer generations of equipment even before the end-of-life to optimize the usefulness and efficiency of available technologies. The grouped model clusters equipment generations and all electricity consumption. This approach is most useful for broadly disseminated environmental impact communication and reporting.

2.2. System Boundary and Assumptions

The system boundary of this LCA considers the raw material extraction and assembly of active operational IT equipment (assets), and their electricity consumption during

operation in the US over the course of 5 years, which is the approximate lifetime of most IT equipment (Associate Director of Environmental Management, personal communication, 26 March 2024). For all equipment, the end-of-life is already built into the Ecoinvent process datasets and is not analyzed separately to avoid the crediting of recycled materials or refurbished equipment (double counting). Data centers have a number of management approaches; in this case, it is an enterprise data center. The enterprise supplies their own white-space assets which draw power from the building gray-space equipment and infrastructure (e.g., air conditioning units, power distribution systems). In this case, the assets represent a portion of the total operations occurring within the same building with a power usage effectiveness (PUE) of 1.88 (Associate Director of Environmental Management, personal communication, 20 September 2024).

The functional unit chosen for this study is 1 kilowatt-hour (kWh) of IT equipment power consumption. This LCA considers multiple generations of technology to represent the necessary rotation of equipment necessary to meet continuous data receipt, storage and access. We assume that newer generations of technology are a lighter weight and more energy-efficient as compared to the older generations; however, the ratio of material inputs remain the same for each technology group. As per feedback from members of the environmental management team at a multinational IT infrastructure services provider, we assume the data center has 20% first generation, 60% second generation, and 20% third generation IT equipment (Associate Director of Environmental Management, personal communication, 17 July 2024).

Not included in the system boundary are power generation or delivery systems (e.g., uninterruptible power supply, backup generators), cooling systems (e.g., computer room air conditioning units, direct expansion air handler units), nor software (e.g., user licenses, integrated applications). Additionally, the physical building structure as well as general services associated with a place of business (e.g., temperature control, lighting, plumbing, operators/staff) are outside of the scope of this project. While the building is outside of the scope of this research, its dimensions are considered for the scenario analysis as described in Section 2.4. This assessment focuses exclusively on IT equipment and its associated electricity consumption, excluding supporting infrastructure and facility-level systems that contribute to overall data center PUE. Therefore, the results should be interpreted as a partial but detailed evaluation of IT-related environmental impacts rather than a complete facility-wide assessment.

2.3. Life Cycle Inventory

All process data was acquired from Ecoinvent version 3.11 (allocation, cut-off by classification—unit) database, which is considered the most comprehensive and well-documented database for region-specific LCA. Input data for the life cycle inventory was informed through communication with the Associate Director of Environmental Management at a multinational information technology infrastructure services provider, who provided a list of sample assets (i.e., physical devices, components, and/or equipment) per group and subgroup (Table 2), weight of the equipment, and nameplate kilowatt (kW) load (Tables S4 and S5). The nameplate kW refers to the rated power of a device, indicating the maximum power the device can handle under normal operating conditions. For active devices, we assume that these are operating on 60% of the kW nameplate capacity, which is a generally accepted assumption among data center management experts as per feedback from the Associate Director of Environmental Management at a multinational IT infrastructure services provider (Associate Director of Environmental Management, personal communication, 20 September 2024). To calculate the electricity consumption, we assume

24–7 operating time over 5 years, and assume the data center was drawing upon medium voltage electricity from the US grid.

Table 2. Summary of IT equipment within sample data center.

Equipment	Definition	Count
Chassis	A hardware container that houses and protects its internal components, and provides them with power, cooling, and connectivity.	272
KVM Switch	Hardware that allows a single keyboard, monitor, and mouse to control multiple computers or servers. KVM stands for “Keyboard, Video (monitor), Mouse.”	18
Network		
Module	Hardware that connects devices to a network that can adapt and expand to meet fluctuating network needs.	647
Standard	Hardware that is integral to communication between devices multiple networks, allowing them to communicate and use the same internet connection.	1601
Peripheral		
Keyboard	An input device that allows users to enter characters and functions into an electronic device.	6
Monitor	An output device that displays information. Also known as a video display unit.	43
Storage	An external device that connects to a computer to enhance its functionality and store data.	785
Server		
Blade	A compact, modular server that manages and distributes data in a network. Also known as high-density servers.	359
Standard	A computer or device that provides resources to other computers of clients over a network.	747

The material breakdown for each asset was informed using the OpenAI large language model (LLM) AI program (OpenAI, GPT-4, queried October 2024) [36,37]. Prompts requested percentage-by-mass material composition for major material categories (e.g., steel, aluminum, copper) for representative enterprise data center equipment types (e.g., servers, storage): Provide an estimated material composition by mass (in percentage terms) for a typical enterprise-grade [chassis/KVM switch/etc.] used in data center environments. Include major material categories such as steel, aluminum, copper, plastics, printed circuit boards, and other electronic components. Provide results as a plausible range for each category. The LLM returned approximate percentage composition for each equipment category (e.g., copper: 8–12%); midpoint values were selected and normalized such that the total composition equaled 100%. The researchers compared these results to publicly available resources, including peer-reviewed LCA sources and industry reports [38–43]. To improve reliability, the generated outputs were reviewed and validated by the multinational IT services provider with expertise in enterprise data center equipment procurement and environmental management (Associate Director of Environmental Management, personal communication, 17 September 2024). An example of this generation and validation process is provided in the Supplementary Material (S2. Life Cycle Inventory). Using the multinational information technology infrastructure services provider data, the total equipment weight and electricity consumption per equipment group was calculated. To create the multiple generations of equipment, we assumed the second generation

(G2) met the average weight and efficiency; the first generation (G1) is 20% heavier and 20% less efficient than G2, and the third generation (G3) is 20% lighter and 20% more efficient than G2 (Associate Director of Environmental Management, personal communication, 17 July 2024). The equipment group weight, lifetime electricity consumption, and material breakdown per kilogram (kg) of equipment type, and Ecoinvent data source can be found in Table S2.

2.4. Sensitivity and Uncertainty Analysis

Sensitivity analysis is a key component of LCA validation which allows us to test the reliability of study results, particularly those that rely on assumptions in the model. A one-at-a-time (OAT) sensitivity analysis was conducted to evaluate the influence of key operational assumptions on life cycle impact results. In this approach, individual parameters are varied independently while all other variables are held constant at their baseline values, allowing for the isolation of each parameter's effect on model outputs. Specifically, IT equipment electricity consumption rates for each equipment generation were varied by $\pm 20\%$ relative to baseline values: G1 (48%), G2 (60%), and G3 (72%). This range was selected to reflect plausible variability in real-world data center operations. For each variation, life cycle impacts were recalculated while maintaining constant values for electricity emission factors, equipment composition, and generation mix. This sensitivity analysis focuses on the midpoint impact factor, global warming potential (kg CO₂ eq).

Uncertainty in the LCA results was evaluated using Monte Carlo simulation on the baseline granular model. A series of stochastic simulations (1000 iterations) was performed, with input values randomly sampled from their respective distributions in each iteration. The resulting impact distributions were used to calculate mean values, standard deviations, and coefficients of variation (CVs) for each impact category. This approach enables a probabilistic assessment of result stability and supports identification of parameters that most influence outcome variability.

2.5. Scenario Analysis

In prior studies, electricity consumption is credited with the majority (60–98%) contribution to overall data center environmental impact [9,11]. The scenario analysis explores how electricity generation mix could impact the performance of IT equipment operations through (1) the use of rooftop solar panels, and (2) a data center operating on a “green” grid (Table S3).

The data center building is two stories with about 6700 square meters (sq.m.) of raised space. The rooftop is about 3300 sq.m.; we calculated the number of 3 kW photovoltaic solar multicrystalline silicon (multi-Si) panel systems that could be installed. A standard 3 kW solar panel system typically requires about 13–18 sq.m., and consists of about 8–10 panels (each generating 300–400 Wp). Using an average of 15 sq.m. per system, we estimated about 220 systems could fit on the building's roof. The total capacity of this system would be 660 kW; however, if we assume 5 h of full sun per day, the system could generate approximately 6 gigawatt-hours (GWh) of electricity per 5 years. The assets consume approximately 63.9 GWh of electricity within their lifetime; if we assume all of the solar electricity generated is used directly by the enterprise, this could displace reliance on the grid electricity generation mix by 10%. We use this value as a consideration for a scenario for on-site electricity generation to improve the data center's reliance on off-site fossil-generated electricity and improve the overall metrics of electricity consumption.

The “green” grid scenario is based on a country's electricity generation mix. Unlike other energy commodities, electricity trade between countries is relatively limited which makes this a good comparison to see how reducing reliance on fossil-generated electricity

could impact overall performance. This study's baseline model uses US electricity data. To compare the performance of the data center IT equipment operating on an electricity grid which relies more heavily on renewable energy sources, we chose Norway as an illustrative example. Ecoinvent 3.11 electricity by country mix is based on 2022 IEA data, which is summarized below in Table 3.

Table 3. Electricity generation mix by country.

Source	United States	Norway
Coal	16.8%	0.1%
Oil	0.7%	0.0% *
Natural gas	41.9%	1.0%
Nuclear	18.2%	n/a *
Hydro	5.9%	89.1%
Biofuel	1.0%	0.0% *
Waste	0.4%	0.3%
Wind	9.7%	9.0%
Solar PV	4.8%	0.2%
Geothermal	0.4%	n/a *
Other sources	0.2%	0.3%

* Note: The sources marked 0.0% indicate a small fraction of the overall electricity generation. The sources marked n/a indicate this source of electricity generation is not used in this country. Source: <https://IEA.org/countries> (accessed on 8 October 2024).

2.6. GHG Corporate Standards Accounting

We calculated emissions using the GHG corporate standards accounting approach and the average-data and spend-based methods to empirically compare the results with the LCA. The IT equipment asset list provided included key information such as asset group, asset identification (often including both brand and parts number), asset weight, asset nameplate, and number of assets. The equipment costs were not included, leading the research team to work backwards to replicate the average-data and spend-based methods. Leveraging the LCI, Ecoinvent database, and LCA output, we created a workbook to calculate kg CO₂ eq for each of these methods. To do this, we analyzed each Ecoinvent process within each equipment group and recorded the cost (EUR2005/kg) and kg CO₂ eq. Using these values, we calculated the cost in 2025 US dollars (USD), kg CO₂ eq per kg, and kg CO₂ eq per USD, which were key for emission estimations using these other methods. For better geographic accuracy, we used the 2024 average price of electricity for commercial utility customers (\$0.1285) sourced from the US EIA [44]. We considered this case study a commercial customer rather than an industrial customer due to the nature of the data center enterprise role in providing services to multiple customers. Because the three GHG accounting methods used in this study rely on a shared inventory (Ecoinvent 3.11), the comparison reflects methodological differences rather than fully independent estimates of GHG emissions.

The average-data method uses the equipment mass (kg) and lifetime electricity consumption (kWh) and multiplies them by secondary cradle-to-gate emission factors. We relied on the AI-informed material breakdown for the mass, and used the kg CO₂ eq per kg emission factor for each Ecoinvent process to calculate the total emissions per equipment category, and summed the findings to reach the total equipment emissions. We then added this to the calculated total emissions associated with electricity consumption to get the overall result.

The spend-based method relies on collecting data on the economic value of purchased goods and services, and multiplying them by an emission factor. We leveraged the provided asset list to perform an internet search for each asset line item to collect the approximate cost. We relied on the AI-informed material breakdown and used the €(EUR2005)/kg (in 2005 Euros) Ecoinvent factor to calculate cost per material. We then adjusted this value to represent 2025 US dollars by adjusting for exchange rate (1.08) and time (1.4) based on present and historical data. Using the Ecoinvent emission factor from the average-data method, and dividing it by the calculated 2025 US dollars, we created a new emission factor kg CO₂ eq per \$. Cost data was largely sourced from brand-provided price lists (e.g., Cisco Global Price List) and online IT equipment retailers (e.g., itprice.com, networkoutlet.com). In some cases, equipment codes on the provided asset list were since discontinued; here we relied on resale sites (e.g., ebay.com) or estimated prices based on similar replacement equipment. Only 80 units (2% of the total equipment) were excluded from the analysis as the label information was too obscure, or we could not find a comparable replacement unit price. Once the cost data was collected, we then multiplied the unit price by the number of units by the emission factor to reach the total equipment emissions. Likewise, this was added to the calculated electricity emissions to reach the final result.

3. Results and Discussion

3.1. Contribution Analysis

Inventory analysis results in a list of hundreds of substance emissions to soil, water, and air. This data is aggregated based on the ReCiPe 2016 method into 18 midpoint impact categories and three endpoint impact categories. For each impact category, the contribution of each process is displayed as a proportion of 100%. Contribution analysis allows for “hotspot” identification across multiple impact factors, which is particularly useful when an LCA model contains several processes as seen in these models. The results from the grouped model and the granular model are equivalent, though the granular model highlights the specific differences between the multiple generations of equipment in terms of equipment impact and electricity consumption impact. The results of these models are summarized in Tables S6 and S7.

Carbon dioxide is the most commonly reported emission as it is the most easily measurable and is most often used in discussion surrounding environmental impact or footprint. For 1 kWh of IT power consumed, 0.710 kg CO₂ eq (kilograms carbon dioxide equivalent) are produced, represented by the impact factor “global warming”. As anticipated, the contribution analysis shows that 67.3% (0.478 kg CO₂ eq) is attributed to electricity consumption (Figure 1 and Table S6). Standard servers had the next highest impact at 20.9% (0.149 kg CO₂ eq), which is largely attributed to the integrated circuit that requires a large amount of electricity and water to produce in addition to precious metals like gold. At a granular level, we learn that the G2 standard server has the overall highest global warming impact, where electricity consumption and the equipment itself are associated with 16.7% and 12.6% of overall CO₂ eq emissions, respectively (Figure 2 and Table S7). Because G2 makes up the largest percentage (60%) of each equipment group, this is not surprising. This result, however, identifies a need for IT equipment developers to focus on the design and operational efficiency of servers in order to improve overall environmental performance and decrease CO₂ eq emissions. When comparing the “blade” and “standard” servers per kWh, the blade server (0.00921 kg CO₂ eq) outperforms the standard server (0.149 kg CO₂ eq) by 93.8%. Transitioning away from standard servers and towards the smaller, more efficient blade server should be explored to reduce overall impact; however, performance-normalized comparisons are needed as the server types are not necessarily interchangeable due to differences in services delivered and storage capacity, among other

factors. Because corporate reporting requires CO₂ emissions estimates, we report our findings using the same; however, more details across impact categories can be found in Figures 1–3 and the Supplementary Material.

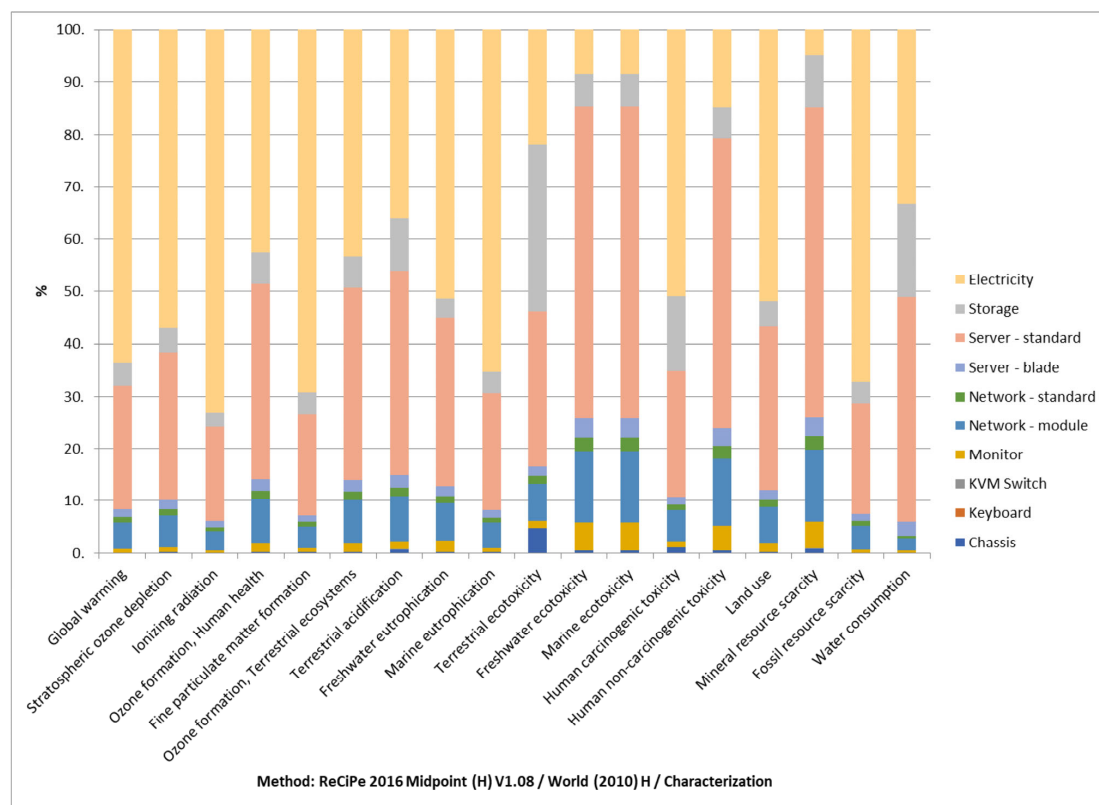


Figure 1. Contribution analysis—midpoint—grouped model.

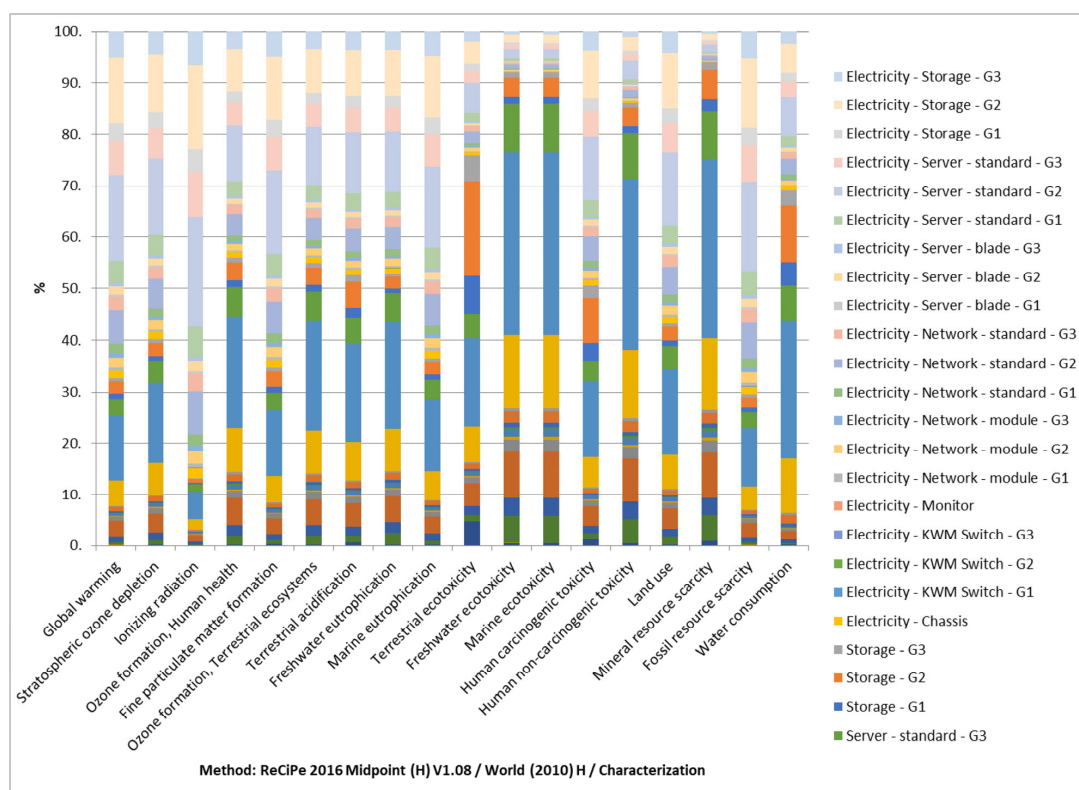


Figure 2. Contribution analysis—midpoint—granular model.

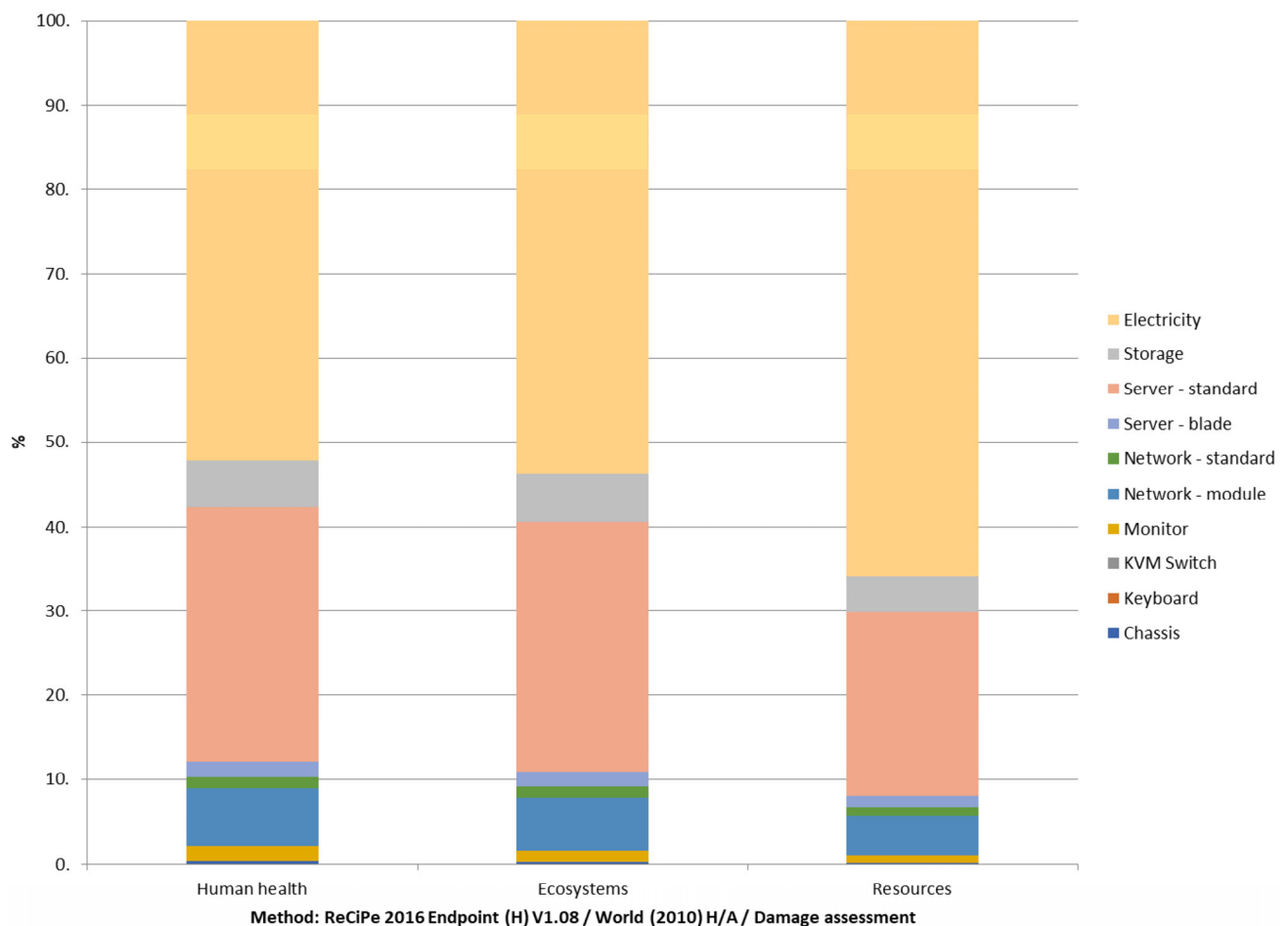


Figure 3. Contribution analysis—endpoint—grouped model.

This trend of electricity consumption having a major impact reigns true across most other midpoint impact categories, indicating that electricity consumption remains a primary source of concern when attempting to reduce overall impact of data centers. However, the standard server equipment has the largest impact among freshwater ecotoxicity (0.0974 kg 1,4-DCB [1,4 dichlorobenzene]), marine ecotoxicity (0.128 kg 1,4-DCB), human non-carcinogenic toxicity (1.53 kg 1,4-DCB), and mineral resource scarcity (0.00497 kg Cu eq [Copper]), which is primarily (75–98%) attributed to the integrated circuit, and water consumption (0.00438 m³ [cubic meters]) which is primarily (79%) attributed to electronic grade silicon.

The granular model provides further details, most notably the generation of each equipment group and the associated electricity consumption. While the model output figures are useful for observing trends, the granular data is more useful for data center managers and decision makers. For example, by adapting the dataset to be more representative of current equipment efficiency vs. future generations of equipment's efficiency, a manager could use this baseline model to inform decisions of when to transition to more efficient technology even before the useful lifetime of the equipment. If CO₂ emissions are the highest priority for the manager, they may decide that efficiency (reduced electricity consumption) must be prioritized over material waste. Alternatively, if moving away from traditional carbon reporting methods, the manager may make the argument for refurbishing older generations of equipment rather than buying new in order to reduce environmental impact on other categories (e.g., ecotoxicity). The endpoint results further aggregate the emissions data into three broad categories: human health, ecosystem quality,

and resource scarcity (Figure 3 and Table S7). The endpoint categories are useful for impact communication. Human health (DALY [disability adjusted life years]) represents the years that are lost or that a person is disabled due to disease or accident, ecosystem quality (species.yr [species per cubic meter per year]) represents the local relative species loss integrated over space and time, and resource scarcity (USD2013 [United States 2013 Dollars]) represents the extra costs involved for future mineral and fossil resource extraction. The results show an impact of 2.36×10^{-6} DALY, 3.37×10^{-9} species.yr, and \$0.04663 associated with the functional unit of 1 kWh of power consumed. As expected from the midpoint model, electricity consumption was the largest contributor across each impact category at 51% (1.2×10^{-6} DALY), 57% (1.93×10^{-9} species.yr), and 70% (\$0.032793), respectively. The standard server also held a notable impact at 31% (7.24×10^{-7} DALY), 27% (9.14×10^{-10} species.yr), and 19% (\$0.008717), respectively.

These results underscore the critical importance of reducing electricity consumption not only within data centers but also throughout their supply chains. Furthermore, ensuring long-term sustainability requires data center managers to account for the material impacts of IT equipment production. Often underestimated, Scope 3 emissions emerge as a significant contributor to the overall environmental footprint. This analysis highlights the need for data centers and their supply chains to internalize these externalities in order to advance meaningful sustainability improvements.

3.2. Sensitivity Analysis

The OAT sensitivity analysis reveals a consistent increase in electricity consumption across all equipment types and generations as utilization rates rise from the low to high scenarios (Table 4). Moving from the low utilization case (38–58%) to the baseline (48–72%) and high scenario (58–86%) results in an approximate 25% increase in electricity consumption across all components, reflecting the proportional scaling embedded in the utilization assumptions. Further, we see a linear trend with percent contribution where the low utilization case contributes to 62.2% of overall kg CO₂ eq emissions, baseline to 67.3% as previously reported, and high utilization to 71.2%. This sensitivity analysis confirms that even among variable electricity utilization rates, electricity consumption remains the greatest contributor to GHG emissions.

Across all the scenarios, server and storage systems dominate total electricity consumption, with standard servers (particularly G2) representing the single largest contribution. Under baseline conditions, G2 standard servers account for 0.118 kg CO₂, substantially exceeding all other components, followed by storage systems (0.091 kg CO₂ eq) and network-standard equipment (0.046 kg CO₂ eq). This pattern persists across low and high scenarios, indicating that a mid-generation unit of equipment operating at moderate utilization levels drives the majority of electricity demand in the system. Generation-level differences are also evident; across all scenarios, G2 represents the largest share of equipment and consistently exhibits the highest electricity consumption. Although G3 operates at higher utilization rates, its overall contribution remains lower due to its smaller proportional representation and improved efficiency characteristics. Peripheral equipment contributes negligibly to total electricity consumption, indicating that efforts to reduce electricity demand should prioritize core IT systems rather than auxiliary components.

The sensitivity results demonstrate that electricity consumption scales predictably with utilization assumptions, with no nonlinear or disproportionate effects observed across equipment types. This suggests that the model is structurally stable and that conclusions regarding dominant contributors are robust to plausible variations in efficiency and use rates.

Table 4. Sensitivity analysis of IT equipment utilization.

Electricity Consumption *									
	Low			Baseline			High		
	G1 (38%)	G2 (48%)	G3 (58%)	G1 (48%)	G2 (60%)	G3 (72%)	G1 (58%)	G2 (72%)	G3 (86%)
Chassis	8.48×10^{-3}			1.06×10^{-2}			1.27×10^{-2}		
KVM Switch									
G1	3.35×10^{-5}			4.19×10^{-5}			5.03×10^{-5}		
G2	8.40×10^{-5}			1.05×10^{-4}			1.26×10^{-4}		
G3	2.23×10^{-5}			2.79×10^{-5}			3.35×10^{-5}		
Monitor	2.08×10^{-4}			2.60×10^{-4}			3.12×10^{-4}		
Network—Module									
G1	4.54×10^{-3}			5.67×10^{-3}			6.80×10^{-3}		
G2	1.14×10^{-2}			1.42×10^{-2}			1.70×10^{-2}		
G3	3.02×10^{-3}			3.78×10^{-3}			4.54×10^{-3}		
Network—Standard									
G1	1.48×10^{-2}			1.85×10^{-2}			2.22×10^{-2}		
G2	3.70×10^{-2}			4.62×10^{-2}			5.54×10^{-2}		
G3	9.84×10^{-3}			1.23×10^{-2}			1.48×10^{-2}		
Server—Blade									
G1	3.51×10^{-3}			4.39×10^{-3}			5.27×10^{-3}		
G2	8.80×10^{-3}			1.10×10^{-2}			1.32×10^{-2}		
G3	2.34×10^{-3}			2.93×10^{-3}			3.52×10^{-3}		
Server—Standard									
G1	3.79×10^{-2}			4.74×10^{-2}			5.69×10^{-2}		
G2	9.44×10^{-2}			1.18×10^{-1}			1.42×10^{-1}		
G3	2.53×10^{-2}			3.16×10^{-2}			3.79×10^{-2}		
Storage									
G1	2.90×10^{-2}			3.62×10^{-2}			4.34×10^{-2}		
G2	7.24×10^{-2}			9.05×10^{-2}			1.09×10^{-1}		
G3	1.93×10^{-2}			2.41×10^{-2}			2.89×10^{-2}		
Totals	3.82×10^{-1}			4.78×10^{-1}			5.73×10^{-1}		
Contribution to overall impact	62.2%			67.3%			71.2%		

* Note: all reported values are for the global warming impact category, kg CO₂ eq, per functional unit 1 kWh.

3.3. Monte Carlo Simulation

A Monte Carlo simulation was conducted on the granular model only, as the grouped model represents a linear aggregation of the same underlying life cycle inventory data. Both models share identical foreground and background inventories and differ solely in the level of data clustering; uncertainty propagation through Monte Carlo simulation yields equivalent impact distributions for both representations. Performing the simulation at the granular level preserves the variability associated with the different generations of IT equipment and operational parameters while avoiding redundant computation. The

CV is a normalized measure of dispersion that describes the relative magnitude of uncertainty in a model output, with a CV < 30% considered generally reliable for interpretation. The results show that the majority of the impact categories meet this standard of reliability, including global warming with a CV of 8.08% (Table S8). Impact categories with a CV $\geq$ 30% are not necessarily invalid, but should be interpreted with caution. To improve the model's results, additional primary emissions data and research linking those emissions with environmental outcomes are needed.

The markedly higher CVs observed for the human health and ecosystem damage endpoints, compared to the relatively low variability for the resource scarcity endpoint, are consistent with the structure of endpoint LCA methods. Human health and ecosystem endpoints are particularly sensitive to uncertainty because they integrate several midpoints characterized by high spatial and temporal variability; small variations at the midpoint level can therefore amplify when translated into endpoint metrics. In contrast, resource scarcity endpoint is dominated by a narrower set of midpoints leading to comparatively stable and lower variability outcomes. While endpoint methodologies are useful for communicating impacts to a wider population, the results underscore the need for cautious interpretation of endpoint results relative to more robust midpoint indicators.

3.4. Scenario Analysis

As anticipated, across most impact categories the baseline scenario performs the lowest (Figure 4 and Table S9). Exceptions include fine particulate matter formation, freshwater eutrophication, terrestrial ecotoxicity, freshwater ecotoxicity, marine ecotoxicity, and human non-carcinogenic toxicity where the rooftop solar scenario performs the lowest. This is primarily due to the physical addition of multi-Si PV solar panels to the system boundary, specifically, the multi-Si PV wafer cells themselves. Further, the “green” grid scenario shows the highest impact in the water consumption category. This is not surprising, as Norway relies heavily on hydropower for its electricity generation.

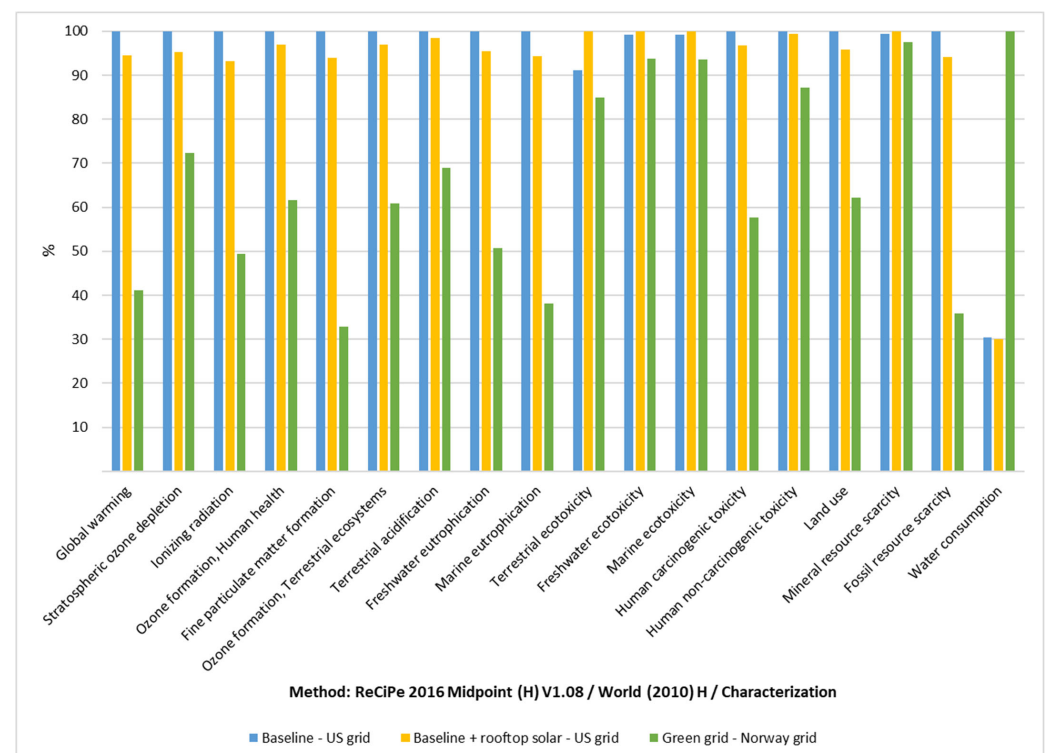


Figure 4. Scenario analysis—comparing baseline results with (1) displaced electricity consumption via solar panel energy generation, and (2) operating on a “green” grid—midpoint.

When observing the global warming impact, higher reliance on renewable energy sources is superior. The rooftop solar scenario displaces approximately 10% of the IT equipment's grid electricity consumption, showing a 15.9% reduction in CO₂ emissions (from 0.710 to 0.597 kg CO₂ eq). This disproportionate reduction occurs because the displaced electricity is more carbon intensive than the system average, while solar energy has a much lower life cycle GHG intensity. Additionally, grid electricity carries upstream emissions from fuel extraction, processing, and combustion that are largely absent for solar PV, whose life cycle impacts are concentrated in manufacturing and amortized over long operational lifetimes. The "green" grid scenario shows a 63.4% reduction (to 0.260 kg CO₂ eq) from the baseline scenario. This supports the evidence that transitioning to a grid with higher reliance on renewable energy sources like hydropower will greatly reduce CO₂ emissions and improve data center performance. When normalizing data, toxicity-related impact categories often emerge as high-priority concerns as they directly affect human and ecosystem health. This highlights the critical need to evaluate the environmental implications of renewable energy technologies comprehensively, rather than just focusing on CO₂ emissions.

While rooftop solar panels reduce CO₂ emissions, the environmental impact in other areas—such as toxicity-related impacts—may outweigh the CO₂ reduction benefits, necessitating a more balanced assessment of their overall sustainability. Further when considering sustainable management of data centers, the financial cost of installation and maintenance to support the benefit of CO₂ reduction might be prohibitive. Our results suggest that the addition of solar panels or operating on a "green" grid does not displace the weight of the environmental impact of standard servers. The results suggest that the standard server equipment must be prioritized as an area of intervention to improve environmental performance beyond consideration of CO₂ emissions. From a practical standpoint, it is also important to consider the long-term benefits of on- and off-site renewable energy generation; while the lifetime of IT equipment modeled in this study is 5 years, solar panels may last up to 30 years and hydropower plants have an infrastructure lifespan of 100+ years. To illustrate these longer-term benefits, a 5-year projection of rooftop solar implementation was conducted. The results show a cumulative reduction of 7645 metric tons kg CO₂ eq compared to the baseline scenario, which is equivalent to the carbon sequestered by 7668 acres of forests in one year according to the US Environmental Protection Agency (EPA) greenhouse gas equivalencies calculator.

The transition to renewable energy technologies is often critiqued as resource-intensive. Building large-scale renewable energy systems like solar and wind farms requires a significant amount of raw materials like metals, rare earth elements, and concrete, which necessitates extensive mining and manufacturing processes. These added inputs could potentially lead to unintended environmental consequences and supply chain issues, disrupting the economy.

3.5. GHG Corporate Standards Accounting

The GHG Protocol defines four general methodologies for corporate carbon accounting methods as detailed in Section 1.2. While LCA is not one of these, it loosely falls under the "hybrid" method which requires knowing an emission factor—in this case, sourced from the Ecoinvent processes of materials needed to produce the IT equipment—and supplier data. The hybrid method defines supplier data as the emissions from electricity and fuel use (which we calculated from nameplate values), mass of material inputs (mass known, material breakdown estimated as detailed in Section 2.3), and waste/emissions output (calculated using Ecoinvent data).

All three methods used in this study rely on Ecoinvent process emissions data; however, the way each method differs in their accounting approach highlights the accuracy of their results (Table 5). When comparing the results, we see that the average-data method is relatively close to the more accurate LCA model results with a difference of only 1.81%. This difference is attributed to the average-data method not accounting for different generations of equipment that were represented in the LCA model. The spend-based method, however, is grossly inflated when compared to the other results—a difference of 40.4% and 38.7% from the LCA model and average-data method results, respectively. This inflation is attributed to two key aspects. First, the emission factor is based on the material makeup of each equipment type, not the equipment itself. Second, the price of IT equipment is not based on the cost of materials alone, but also the overhead, labor, shipping, and other market factors that influence price.

Table 5. Comparison of three carbon accounting methods.

Category	LCA Model (kg CO ₂ eq per kWh)	Average-Data Method (kg CO ₂ eq per kWh)	Spend-Based Method (kg CO ₂ eq per kWh)
Chassis	7.60×10^{-4}	7.59×10^{-4}	8.20×10^{-2}
KVM Switch	9.26×10^{-5}	9.28×10^{-5}	2.67×10^{-4}
Network—Module	5.67×10^{-3}	5.67×10^{-3}	1.64×10^{-2}
Network—Standard	3.50×10^{-2}	3.50×10^{-2}	2.27×10^{-1}
Keyboard	2.74×10^{-7}	2.94×10^{-7}	4.65×10^{-6}
Monitor	4.12×10^{-3}	4.05×10^{-3}	4.64×10^{-4}
Storage	2.78×10^{-2}	2.78×10^{-2}	2.27×10^{-1}
Server—Blade	9.21×10^{-3}	9.21×10^{-3}	2.36×10^{-3}
Server—Standard	1.49×10^{-1}	1.49×10^{-1}	2.67×10^{-2}
Electricity Consumption	4.78×10^{-1}	4.91×10^{-1}	4.91×10^{-1}
Total	7.10×10^{-1}	7.23×10^{-1}	1.07

These considerations underscore a critical gap in the industry: the lack of supplier-specific cradle-to-gate emission factors for each category of IT equipment type, or ideally, emission factors for each individual product manufactured and sold. Moreover, emission factors in the LCA-based approach extends beyond CO₂ to include other climate forcers and pollutants to enable a more comprehensive assessment of sustainable growth, allowing stakeholders to evaluate trade-offs and co-benefits across broader environmental objectives. Such granular data for data centers is available with the sustainability managers of the companies operating the data centers and following this approach would substantially improve the accuracy of emissions accounting and enable data centers and their suppliers to more precisely identify environmental hotspots across the supply chain. This, in turn, could inform the strategic redesign of IT equipment to optimize functional performance, energy and cost efficiency, and also help data center companies in corporate communications of being more environmentally benign entities.

Many US-based corporations rely on the open access US EPA GHG emission factor database, which provides emission factors for all North American Industry Classification System (NAICS)-defined commodities to perform the spend-based GHG accounting method (Supply Chain Greenhouse Gas Emission Factors v1.3 by NAICS-6, 2024). The datasets provide an emission factor for kg CO₂ eq per 2022 US dollar for 18 separate GHGs as well as a separate average emission factor for all GHGs combined. The datasets are

expensive, accounting for 1016 NAICS commodities. Despite this, these datasets are not nearly as comprehensive as Ecoinvent nor several other private environmental impact datasets used in our LCA model. This limitation makes GHG accounting and reporting less granular, and therefore less accurate—particularly when considering Scope 3 emissions. Further, the emission factors are limited to GHGs. While GHGs are a main concern and focus of most environmental reporting, they are not necessarily indicative of an organization's true environmental impact. Methods like ReCiPe within LCA models allow for a comprehensive approach to environmental impact assessment, and support decision-makers' ability to evaluate trade-offs beyond GHGs to address additional sustainability metrics. As demand for data center capacity continues to grow, improvements in supply chain transparency and product-level emissions data could play a pivotal role in supporting sustainable growth—providing the desired storage and services without overloading electrical grids or exacerbating energy burden concerns for electricity consumers.

4. Conclusions

Data centers are one of the fastest-growing sectors, and are considered to have a high impact on global environmental health. Accordingly, entities operating within this sector are increasingly required to report their environmental impact by estimating their direct and indirect CO₂ emissions. However, research has shown that indirect emissions have been historically underestimated. Further, the large share of indirect value chain emissions can lead to complexities in accounting for the complete footprint of a data center as they require greater monitoring and visibility across supply chain partners and/or require a great deal of assumptions in emission estimation calculations and modeling in the absence of product-level GHG inventory data. To this end, this study applies LCA to a data center enterprise's IT equipment and electricity consumption, and provides a more comprehensive approach to environmental accounting.

The results of the LCA indicate that for every kWh of power consumed by IT equipment, approximately 0.710 kg CO₂ eq was emitted. Of the total emissions, 67.3% is directly attributed to the operation (electricity consumption) of IT equipment, thereby demonstrating the dominance of the operational phase in the life cycle impact of the IT equipment. The remaining CO₂ eq emissions stem from the indirect supply chain emissions of the IT equipment. Standard servers, in particular, accounted for 20.9% of these indirect emissions. The granularity provided by this LCA can be useful in understanding the embodied impact from various phases including the raw material extraction, manufacturing processes, and end-of-life disposal of IT components. Sensitivity and uncertainty analyses validate the model and confirm the study's core findings. While utilization assumptions influence absolute electricity consumption, they do not alter the relative importance of equipment categories or the dominance of electricity use as a key driver of CO₂ eq emissions. Further, uncertainty analysis identifies key impact categories which would benefit from additional primary data while also revealing high certainty levels across several impact categories, including CO₂ eq emissions.

Scenario analysis was done to explore the implications of a clean energy transition. The results indicated that the rooftop solar scenario resulted in a 15.9% reduction in CO₂ eq emissions, whereas the green grid scenario showed a 63.4% reduction in CO₂ eq emissions. Technology advancements to improve recycling capabilities, developing sustainable mining practices, and research into the use of new materials and manufacturing techniques should continue to be explored so that the transition to a "green" grid does not come with unintended environmental consequences beyond its impact on CO₂ emissions during the use phase.

GHG corporate accounting approaches remain the more popular method in emission calculation and reporting, primarily due to the non-compilation of product-level GHG data and relative simplicity. However, the level of accuracy of these approaches is questionable, yet not fully understood. To address this gap and provide an empirical comparison of these approaches with LCA, we apply the GHG corporate accounting approach using average-data and spend-based methods. The average-data method produced comparable results at 0.723 kg CO₂ eq per kWh. When the underlying emission factor is reliable, this average-based approach can provide reasonable results, even though it does not capture differences in efficiency across equipment generations. In contrast, the spend-based method resulted in a substantial overestimation of emissions at 1.07 kg CO₂ eq per kWh, underscoring its limited reliability for data-informed interventions. This overestimation arises largely because expenditures reflect numerous factors beyond material inputs, including overhead, labor, transportation, and other indirect costs. This highlights the importance of avoiding spend-based methods for decision-critical applications—such as carbon pricing or product-level footprinting—unless no other data is available.

Together, these findings reinforce the importance of collaboration with supply chain partners to collect and share product-level data and emission factors to improve the accuracy, consistency, and transparency of emissions accounting and reporting. The LCA-based approach addresses many of these shortfalls by offering a more comprehensive and scientifically robust framework for environmental accounting. LCA evaluates environmental impacts across the full life cycle using process-specific data that reflect the actual flows of materials and energy. The system boundary transparency and methodological rigor enable the LCA-based approach to capture upstream and downstream emissions that corporate methods often overlook, such as emissions embodied in the supply chain or those associated with product disposal. Moreover, the LCA-based approach goes beyond GHG emissions to consider multiple environmental impact categories, offering a more holistic assessment aligned with sustainability goals.

These methodological differences carry important policy implications, particularly for high energy users such as data centers, where emission profiles are complex, rapidly evolving, and highly sensitive to system boundaries and accounting choices. The current lack of standardization across GHG accounting methods can lead to inconsistent reporting, reduced comparability, and potential misalignment between reported emissions and actual systems-level impacts. This creates challenges for regulators, investors, and other stakeholders seeking to benchmark performance, design incentives, or enforce emissions reduction targets. Standardizing key elements of accounting methodologies could improve comparability and credibility across firms while reducing opportunities for selective reporting. At the same time, policy frameworks should recognize the trade-offs between simplified corporate accounting methods and more data-intensive LCA approaches. While LCA offers greater accuracy and decision relevance, it also requires higher data availability, technical expertise, and computational resources. As such, hybrid approaches or tiered reporting standards may be necessary, where screening-level GHG accounting is complemented by LCA-based analyses for high-impact sectors, processes, or investment decisions. For data centers and other energy-intensive industries, policies that incentivize or require LCA-informed reporting could better align emissions accounting with actual environmental outcomes.

For decision-makers aiming to understand evidence-based trade-offs or improve environmental performance across complex systems, LCA provides a more accurate and actionable foundation than GHG corporate accounting approaches, which tend to oversimplify and aggregate emissions in ways that obscure meaningful intervention points. This underscores the need to develop targeted training and resources to help sustainability

teams, procurement officers, and executives understand the methodological trade-offs and interpret emissions data appropriately such that they can make meaningful abatement decisions. Strengthening both methodological literacy and policy alignment will be critical to ensuring that emissions accounting frameworks drive meaningful and verifiable decarbonization outcomes.

5. Limitations and Future Research

This study is not without limitations and opens up several avenues for future research. The primary data used to validate this framework were drawn from an enterprise data center based in the United States, which limits the generalizability of the results. As demand and services provided by data centers continue to evolve—for example, through the expansion of cloud computing and AI applications—additional frameworks of this kind should be developed to reflect emerging operational and geographical realities. Additionally, the system boundaries of this study were limited to active IT equipment and electricity use. Future research should also more systematically examine the environmental contributions of gray-space equipment and the building itself, including structural, electrical, interior finishing and exterior components, while distinguishing between new construction and retrofits. Further, because all GHG accounting approaches used in this study are derived from a shared inventory, the comparison reflects methodological difference rather than fully independent estimates. To fully evaluate both the accounting methodologies and calculate emissions, future research can replicate this study’s framework using independent data sources for emission factors beyond those used in Ecoinvent. Although IT equipment typically accounts for the largest share of Scope 3 emissions, the broader spectrum of environmental impacts associated with data centers—including other GHGs, resource depletion and localized pollution—remains less understood. Expanding the scope of analysis and improving access to supplier-specific product-level emissions data will be critical for supporting transparent, multi-dimensional evaluations of sustainability performance across the entire data center life cycle.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su18115671/s1>. Table S1: Description of ReCiPe 2016 midpoint and endpoint impact categories; Table S2: Information Technology (IT) Assets; Table S3: Electricity Consumption (Scenarios); Table S4: Grouped Model; Table S5: Granular Model; Table S6: Midpoint and endpoint combined results—functional unit 1kWh—Grouped Model; Table S7: Midpoint and endpoint combined results—functional unit 1 kWh—Granular Model; Table S8: Monte Carlo simulation—midpoint results—Granular Model; Table S9: Scenario analysis—midpoint and endpoint combined results.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
CO ₂	Carbon dioxide
CV	Coefficient of variation
DALY	Disability adjusted life years
EPA	Environmental Protection Agency
EU	European Union
G1	First generation
G2	Second generation
G3	Third generation
GHG	Greenhouse gas
GWh	Gigawatt-hour
IEA	International Energy Agency
IT	Information technology
kg	Kilogram
kg CO ₂ eq	Kilograms carbon dioxide equivalent
kW	Kilowatt
kWh	Kilowatt-hour
LCA	Life cycle assessment
LCI	Life cycle inventory
US	United States
USD	United States dollars

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